



Class : 8<sup>th</sup>  
Name : \_\_\_\_\_  
Section : \_\_\_\_\_  
Marks : 25  
Time : 1h 30m  
Subject : Mathematics

### I. Answer any 3 of the following

(3 x 4 = 12M)

1. In a model of a ship, the mast is 9cm high, while the mast of the actual ship is 12m high. If the length of the ship is 28m, how long is the model ship?

#### 2. Evaluate:

○ i)  $\frac{8^{-1} \times 5^3}{2^{-4}}$

○ ii)  $(5^{-1} \times 2^{-1}) \times 6^{-1}$

3. If a box of sweets is divided among 24 children, they will get 5 sweets each. How many would each get if the number of the children is reduced by 4?

#### 4. Express the following numbers in standard form:

i) 0.000000000085

ii) 6020000000000000

iii) 0.00000000837

iv) 0.0000000000942

### II. Answer the following questions

(4 x 2 = 8M)

#### 5. Evaluate:

○ i)  $5^{-2}$

○ ii)  $(-4)^{-2}$

6. Express the following number in usual form:  $4.05 \times 10^{-6}$

7. Find the value of  $(2^{-1} \times 4^{-1}) \div 5^2$

8. A machine in a soft drink factory fills 840 bottles in six hours. How many bottles will it fill in five hours?

### III. Fill in the blanks

(5 x 1 = 5M)

9.  $a^m \times b^m =$  \_\_\_\_\_

10.  $a^m \times a^n =$  \_\_\_\_\_

11. If  $x_1, x_2$  are the values of 'x' and  $y_1, y_2$  are the values of 'y' then direct proportion is  
\_\_\_\_\_

12. The cost of 5 apples is Rs.100, then the cost of 3 apples is \_\_\_\_\_

13.  $(a^m)^n =$  \_\_\_\_\_